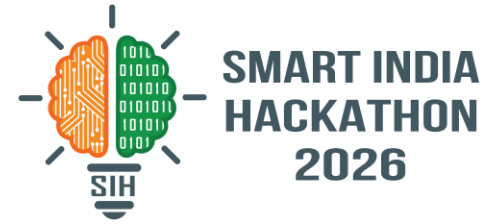
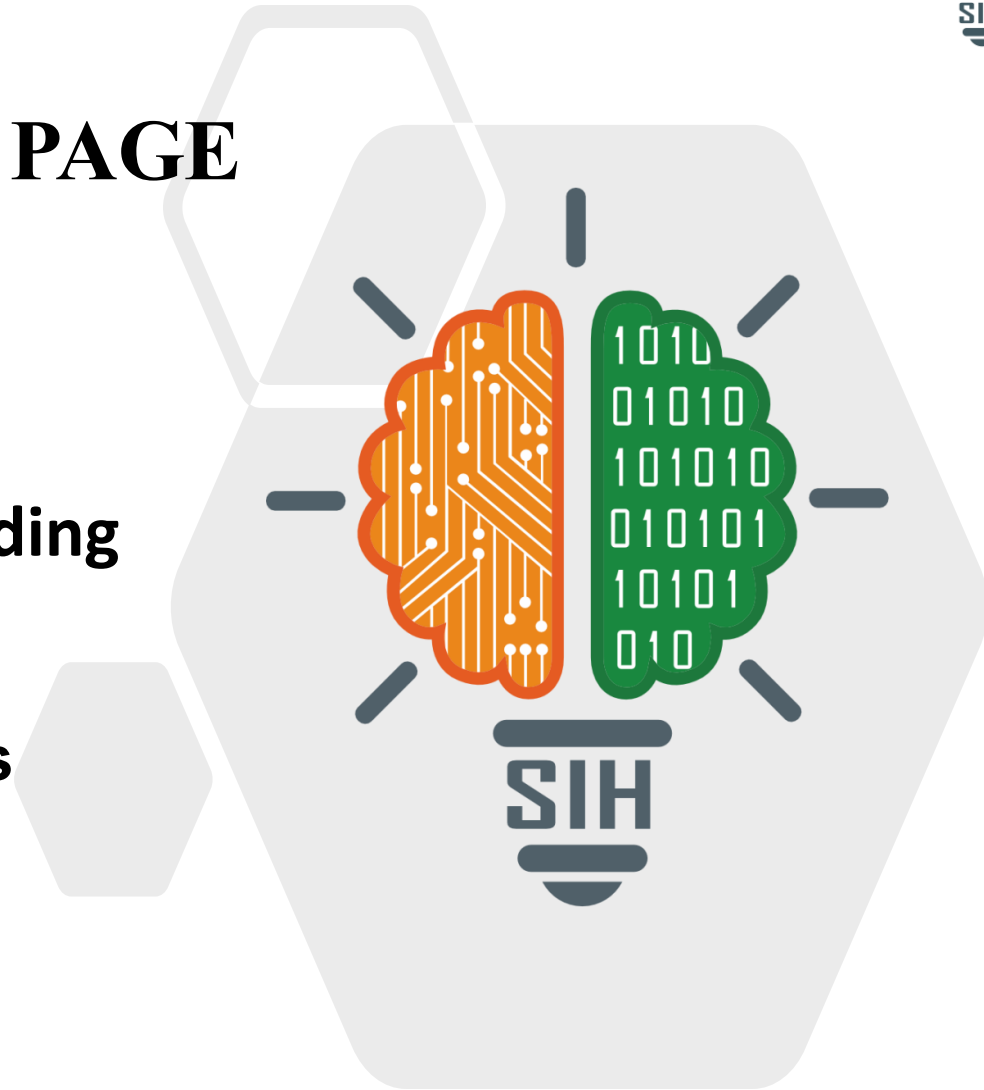


SMART INDIA HACKATHON 2026



TITLE PAGE

- Problem Statement –
Intelligent Public Transport Crowding
& Route Prediction
- Theme-Transportation & Logistics
- PS Category- Software
- Team Name - Medha



Proposed Solution

- Per-stop crowding & delay prediction
- Real-time formulas: time, weather, disruptions
- Cascade model: disruption spreads with distance
- Auto route-finding + best option ranking
- Naive vs recommended comparison
- XGBoost model running alongside rule engine

How it addresses the problem

- Removes commuter guesswork
- Early cascade warning for authorities

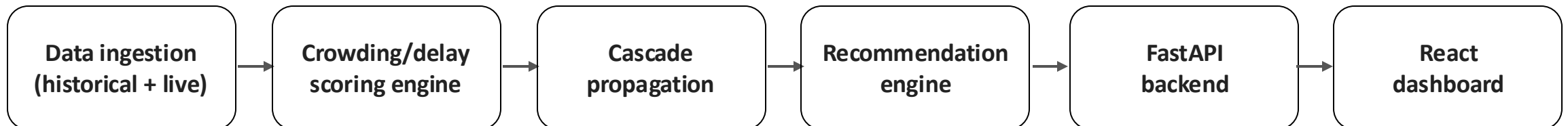
Innovation & uniqueness

- Shows how disruptions spread, not just current status
- Naive-vs-optimized comparison makes benefit tangible
- Lightweight, explainable core — easy to upgrade to ML

Technologies used

- Python (Pandas, NumPy) — data & scoring engine
- XGBoost — trained crowding model, validated on held-out test data
- FastAPI — backend serving predictions
- React + Recharts — live dashboard
- GTFS feed — real route/stop data
- SQLite — lightweight data store
- Open-Meteo — live weather intensity, no API key required
- BMTC GTFS (Vonter/bmtc-gtfs) — real Bengaluru route/stop network

Methodology — process flow



Feasibility

- Uses open GTFS standard — no proprietary data needed
- No training data or GPU required
- Phased rollout: one corridor first, then scale

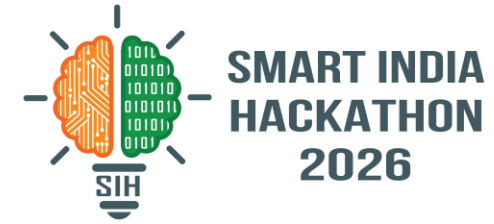
Potential challenges & risks

- Live data feeds vary by transit agency
- Accuracy depends on historical data quality
- Needs integration with an existing transit app

Strategies to overcome them

- Fallback: static schedule + crowdsourced reports
- Phased data-sharing deal with transit authority
- Ship as API/plugin, not a standalone app

IMPACT AND BENEFITS



Potential impact on target audience

- Commuters get clear, actionable route guidance
- Authorities get early warning of disruptions

Benefits

- Social: less commute stress, safer boarding
- Economic: better fleet use, less time lost
- Environmental: shift from private vehicles to transit

- GTFS standard — gtfs.org
- Open transit datasets — data.gov.in
- Internal causal-graph model, adapted from network-fault prediction
- Literature on transit crowding & delay propagation
- XGBoost — trained model docs